

2-Mark Questions & Answers — MPCA Module 5: 80386DX & Pentium Processors

Q1. What are the bus widths and addressable memory of 80386DX?

- **Data Bus:** 32-bit (transfers 4 bytes per cycle).
 - **Address Bus:** 32-bit.
 - **Physical Memory:** $2^{32} = 4 \text{ GB}$.
 - **Virtual Memory:** 64 Terabytes (TB).
-

Q2. List the 3 operating modes of 80386DX.

1. **Real Address Mode:** 16-bit 8086 clone mode (1 MB memory).
 2. **Protected Virtual Address Mode:** 32-bit multitasking mode with 4-level privilege rings (0-3) and Paging.
 3. **Virtual 8086 Mode:** Sub-mode running legacy 8086 apps under protected OS.
-

Q3. What is superscalar operation in Pentium?

Superscalar Execution allows the processor to execute **more than one instruction per clock cycle** by using two parallel integer execution pipelines (**U-Pipe and V-Pipe**).

Q4. What are the four states of the MESI cache protocol?

- **M (Modified):** Dirty line, exclusive to cache, memory is stale.
 - **E (Exclusive):** Clean line, exclusive to cache, matches memory.
 - **S (Shared):** Clean line, present in multiple caches, matches memory.
 - **I (Invalid):** Line is invalid / stale.
-